Chatbot Project Report

Title: Classifying User Intents using DistilBERT-based Chatbot

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1. Introduction

This project aims to build a chatbot capable of classifying user queries into predefined categories using Natural Language Processing (NLP) and Machine Learning techniques. The goal is to develop a functional chatbot that can understand and respond appropriately based on trained data.

The chatbot leverages the CLINC150 "out-of-scope" dataset and uses a fine-tuned DistilBERT model to classify user intents. Additionally, the system includes voice input via speech recognition and a Streamlit web interface for interaction.

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2. Dataset

Source: `clinc_oos` dataset from Hugging Face Datasets **Structure:**

- `text`: User input/question
- `intent`: Category label for the input

The dataset includes 150 in-scope intent classes and one out-of-scope class, making it ideal for intent recognition.

Preprocessing Steps:

- Lowercasing
- Tokenization using `DistilBertTokenizerFast`
- Special characters and stopwords are inherently handled by the tokenizer
- Used Hugging Face `datasets` library for efficient dataset management

Although classical preprocessing like stemming and lemmatization are generally used, we avoided them here since BERT-based tokenizers inherently manage linguistic structure.

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3. Model Selection and Training

Model Used: DistilBERT (`distilbert-base-uncased`) from Hugging Face Transformers
Task: Sequence Classification (Multi-class)

Why DistilBERT?

- Lightweight and efficient compared to BERT
- High accuracy in intent recognition tasks

Training Details:

- 3 epochs
- Batch size: 8
- Learning rate: Default
- Training done using `Trainer` API from Hugging Face

Evaluation Metrics:

- Accuracy: 89%
- Precision, Recall, and F1-score were also computed

The model performed well in recognizing various intents with balanced performance across most categories.

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4. Chatbot Interaction System

Interface: Developed using `Streamlit`
Voice Input: Integrated using `SpeechRecognition` library

Functionality:

- User inputs a query (typed or spoken)
- The query is tokenized and passed to the trained model
- The predicted intent is returned as the chatbot response

The system can classify whether an input is within one of the 150 intents or is out-ofscope.

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5. Evaluation and Testing

Testing:

- The model was tested on a held-out test set
- Random and edge-case inputs were evaluated manually

Results:

- Consistent performance with 89% accuracy
- Model generalized well to unseen examples

Confusion Matrix: Used to analyze misclassifications and verify balanced predictions across classes.

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6. Conclusion and Future Work

This chatbot project demonstrates the effectiveness of transformer-based models like DistilBERT in handling intent classification tasks. By integrating a modern NLP model with an interactive GUI and voice input, the project meets both the technical and user experience objectives.

Future Enhancements:

- Add rule-based fallback for unclear predictions
- Improve accuracy using full BERT or RoBERTa
- Add response generation instead of just intent classification
- Deploy on Streamlit Cloud or Hugging Face Spaces

Attachments:

- Python Script: `chatbot_app.py`
- Evaluation Metrics Screenshot
- Streamlit Interface Demo

Note: Classical ML models (e.g., Logistic Regression, Naive Bayes) were not used in final implementation due to superior performance of transformer models, but could be tested for comparison in future work.

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